

Qiu Yang

Assistant Professor at Peking University (PKU)
Department of Atmospheric and Oceanic Sciences, School of Physics
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<https://qiuyang50.github.io/>
Last update: September 5, 2026

RESEARCH INTERESTS

How strong could mesoscale convective systems and Madden-Julian Oscillation be in the future? What dominant mechanisms underlie these atmospheric phenomenon? My current research focuses on investigating the potential response of convective organization to climate change. Utilizing theoretical models, high-resolution convection-permitting simulations, rigorous analysis of observational data, and AI tools, I aim to elucidate the essential underlying mechanisms of convective organization.

PROFESSIONAL EXPERIENCE

Assistant Professor 2023 Nov.–present

- Address: Department of Atmospheric and Oceanic Sciences, School of Physics, Peking University
- Research Interest: Large-scale convective organization (e.g., MJO), Atmospheric dynamical theory and modeling, High-resolution cloud-resolving simulation

Research Scientist 2020 Sep.–2023 Oct.

- Address: Atmospheric Sciences and Global Change Division, Pacific Northwest National Laboratory (PNNL), USA
- Project: 1) Water Cycle and Climate Extremes Modeling, 2) E3SM Multiscale Modeling Framework
- Project PI: Lai-yung Ruby Leung (PNNL)

Visiting Scholar 2018 Mar.–May

- Climate and Global Dynamics Laboratory, National Center for Atmospheric Research (NCAR), USA
- Host: Mitchell Moncrieff

Visiting Scholar 2017 Sep.–Dec., 2019 May–Jul.

- Department of Mathematics and Statistics, University of Victoria (UVic), Canada
- Host: Boualem Khouider

Postdoc Associate 2017 Jul.–2020 Aug.

- Address: Center for Atmosphere Ocean Science, Courant Institute, New York University, USA
- Project: Upscale Impact of Mesoscale Convective Systems on Tropical Weather and Climate
- Mentor: Andrew Majda (NYU), Mitchell Moncrieff (NCAR), and Boualem Khouider (UVic)

EDUCATION

Ph.D. in Mathematics and Atmosphere & Ocean Sciences 2012 fall–2017 spring

- Center for Atmosphere Ocean Science, Courant Institute, New York University (NYU), NY, USA
- Thesis: Multi-Scale Models for the Scale Interaction of Organized Tropical Convection

· Advisor: Andrew J. Majda (NYU)

B.S. in Mathematics (major) and Physics (minor)

2008 fall–2012 spring

- Zhiyuan College, Shanghai Jiao Tong University (SJTU), Shanghai, China
- Thesis: Optimal Transport of Water in the Biological Network
- Advisor: David Cai (NYU) and Dan Hu (SJTU)

PUBLICATIONS

Below * denotes corresponding author and my name is marked in bold.

2026

- [26] Bai, L., Huang, Y., Liu, Q., Chen, G., Fu, S., **Yang, Q.** and Zhu, L., 2026. Orographic disturbances shaping convective preconditioning over a tropical island. *Atmospheric Research*, <https://doi.org/10.1016/j.atmosres.2026.109252>
- [25] Li, Y. and **Yang, Q.***, 2026. A Two-Layer Stochastic Dynamic Model for Convective Aggregation over Tropical Oceans: Roles of Compensating Subsidence, Cold Pools, and Vertical Shear. *Journal of the Atmospheric Sciences*, <https://doi.org/10.1175/JAS-D-26-0050.1>.
- [24] **Yang, Q.***, Leung, L.R., Li, J., Feng, Z., Wang, H. and Yu, Y., 2026. Observed and Simulated Characteristics of Central US Mesoscale Convective Systems and Their Dependence on Monthly Environmental Conditions. *Journal of Climate*, <https://doi.org/10.1175/JCLI-D-25-0601.1>
- [23] Lin, G., Liao, W., Lin, Z., Zhang, H., Kou, W., Guo, X., Xie, Z., **Yang, Q.**, Wu, C. and Zhang, M., 2026. Development of CAS-ESM-MMF: improving East Asian summer precipitation simulation with a Multiscale Modeling Framework. *Geoscientific Model Development*, <https://doi.org/10.5194/gmd-19-327-2026>.

2025

- [22] Nie, J., Lin, Z., Li, P., **Yang, Q.**, Ding, T., Song, F., Meng, Z., Sun, J., Li, J. and Zhao, C., 2025. Mesoscale convective systems: From interannual and decadal variability to responses to climate change. *Acta Meteorologica Sinica*, <https://dx.doi.org/10.11676/qxxb2025.20240112>.
- [21] **Yang, Q.***, Leung, L.R., Feng, Z., Li, J. and Chen, X., 2025. Environmental thermodynamic effects on future changes of mesoscale convective systems and isolated deep convection over the US: A case study. *Journal of Geophysical Research: Atmospheres*, <https://doi.org/10.1029/2025JD044740>

2023

- [20] Xingchao Chen*, Ruby Leung, Zhe Feng, Xingchao Chen, **Qiu Yang**, 2023: *Environmental controls on MCS's lifetime rainfall over tropical oceans*. *Geophysical Research Letters*, <https://doi.org/10.1029/2023GL103267>
- [19] **Qiu Yang***, L. Ruby Leung, Zhe Feng, Xingchao Chen: *A Moist Potential Vorticity Model for Mid-Latitude Mesoscale Convective Systems over the Land*. *Journal of the Atmospheric Sciences*, <https://doi.org/10.1175/JAS-D-22-0244.1>
Media coverage: DOE EESM program research highlight <https://climatemodeling.science.energy.gov/research-highlights/building-theoretical-dynamical-model-thunderstorms-over-united-states>
- [18] Xingchao Chen*, Ruby Leung, Zhe Feng, Xingchao Chen, **Qiu Yang**, 2023: *Diurnal MCS prior to the genesis of tropical cyclone Mora (2017): the role of convectively forced gravity waves*. *Journal of the Atmospheric Sciences*, <https://doi.org/10.1175/JAS-D-22-0203.1>.

2022

- [17] **Qiu Yang***, L. Ruby Leung, Zhe Feng, Xingchao Chen, 2022: *Impact of Global Warming on U.S. Summertime Mesoscale Convective Systems: A Simple Lagrangian Parcel Model Perspective*. Journal of Climate, <https://doi.org/10.1175/JCLI-D-22-0291.1>
Media coverage: DOE EESM program research highlight <https://climatemodeling.science.energy.gov/research-highlights/using-theoretical-model-predict-future-thunderstorm-changes-over-united-states>
- [16] **Qiu Yang***, Walter Hannah, L. Ruby Leung, 2022: Convective Momentum Transport and its Impact on the Madden-Julian Oscillation in E3SM-MMF. Journal of Advances in Modeling Earth Systems. e2022MS003206, <https://doi.org/10.1029/2022MS003206>
- [15] Fengfei Song*, L. Ruby Leung, Zhe Feng, Xingchao Chen, **Qiu Yang**, 2022: Historical and future changes of large-scale environments for spring MCS initiation over the U.S. Great Plains. Geophysical Research Letters, 49(15), e2022GL098799, <https://doi.org/10.1029/2022GL098799>
- [14] Xingchao Chen*, L. Ruby Leung, Zhe Feng, **Qiu Yang**, 2022: *Precipitation-moisture interactions over tropical oceans: consecutive roles of different convective systems*. Geophysical Research Letters, 49(7), p.e2022GL097836, <https://doi.org/10.1029/2022GL097836>

2021

- [13] **Qiu Yang***, L. Ruby Leung, Zhe Feng, Fengfei Song, Xingchao Chen, 2021: *A Simple Lagrangian Parcel Model for the Initiation of Summer-time MCSs over the Central US*, Journal of the Atmospheric Sciences, 78(11), 3537-3558, <https://doi.org/10.1175/JAS-D-21-0136.1>
Media coverage: DOE EESM program research highlight <https://climatemodeling.science.energy.gov/research-highlights/building-theoretical-model-thunderstorms-over-central-united-states>
- [12] Xingchao Chen*, L. Ruby Leung, Zhe Feng, Fengfei Song, **Qiu Yang**, 2021: *Mesoscale Convective Systems Dominate the Energetics of the South Asian Summer Monsoon Onset*, Geophysical Research Letters, e2021GL094873, <https://doi.org/10.1029/2021GL094873>
- [11] **Qiu Yang***, Andrew J. Majda, Nan Chen, 2021: *ENSO Diversity in a Tropical Stochastic Skeleton Model for the MJO, El Nino, and Dynamic Walker Circulation*, Journal of Climate, 34(9), pp. 3481-3501, <https://doi.org/10.1175/JCLI-D-20-0447.1>

2019

- [10] **Qiu Yang***, Andrew J. Majda, Noah D. Brenowitz, 2019: *Effects of Rotation on the Multiscale Organization of Convection in a Global 2D Cloud Resolving Model*, Journal of the Atmospheric Sciences, 76(11), pp.3669-3696, <https://doi.org/10.1175/JAS-D-19-0041.1>
- [9] **Qiu Yang***, Andrew J. Majda, Mitchell W. Moncrieff, 2019: *Upscale Impact of Mesoscale Convective Systems and Its Parameterization in An Idealized GCM for An MJO Analog above the Equator*, Journal of the Atmospheric Sciences, 76(3), pp.865-892, <https://doi.org/10.1175/JAS-D-18-0260.1>
- [8] **Qiu Yang**, Boualem Khouider*, Andrew J. Majda, Michele De La Chevrotiere, 2019: *Northward Propagation, Initiation and Termination of Boreal Summer Intraseasonal Oscillations in a Zonally Symmetric Model*, Journal of the Atmospheric Sciences, 76(2), pp.639-668, <https://doi.org/10.1175/JAS-D-18-0178.1>
- [7] **Qiu Yang***, Andrew J. Majda, 2019: *Upscale Impact of Mesoscale Disturbances of Tropical Convection on 2-Day Waves*. Journal of the Atmospheric Sciences, 76(1), pp.171-194, <https://doi.org/10.1175/JAS-D-18-0049.1>

2018

- [6] Noah D. Brenowitz*, Andrew J. Majda, **Qiu Yang**, 2018: *The Multiscale Impacts of Organized Convection in Global 2-D Cloud-Resolving Models*. Journal of Advances in Modeling Earth Systems,

10(8), pp.2009-2025, <https://doi.org/10.1029/2018MS001335>

- [5] **Qiu Yang*** and Andrew J. Majda, 2018: *Upscale Impact of Mesoscale Disturbances of Tropical Convection on Convectively Coupled Kelvin Waves*. Journal of the Atmospheric Sciences, 75(1), pp.85-111, <https://doi.org/10.1175/JAS-D-17-0178.1>

2014-2017 (my Ph.D. studies)

- [4] **Qiu Yang***, Andrew J. Majda and Boualem Khouider, 2017: *ITCZ Breakdown and Its Upscale Impact on the Planetary-Scale Circulation over the Eastern Pacific*. Journal of the Atmospheric Sciences, 74(12), pp.4023-4045, <https://doi.org/10.1175/JAS-D-17-0021.1>
- [3] **Qiu Yang*** and Andrew J. Majda, 2017: *Upscale Impact of Mesoscale Disturbances of Tropical Convection on Synoptic-Scale Equatorial Waves in Two-Dimensional Flows*. Journal of the Atmospheric Sciences, 74(9), pp.3099-3120, <https://doi.org/10.1175/JAS-D-17-0068.1>
- [2] Andrew J. Majda and **Qiu Yang***, 2016: *A Multi-Scale Model for the Intraseasonal Impact of the Diurnal Cycle over the Maritime Continent on the Madden-Julian Oscillation*. Journal of the Atmospheric Sciences, 73(2), pp.579-604, <https://doi.org/10.1175/JAS-D-15-0158.1>
- [1] **Qiu Yang*** and Andrew J. Majda, 2014: *A Multi-scale Model for the Intraseasonal Impact of the Diurnal Cycle of Tropical Convection*. Theoretical and Computational Fluid Dynamics, 28(6), pp.605-633, <https://doi.org/10.1007/s00162-014-0336-3>

CONFERENCES AND TALKS

- [40] **06/25/2024 Invited talk** in session AS77 of AOGS 21st annual meeting at Pyeongchang, Gangwon-do, South Korea. The title of my talk is “*Impact of Global Warming on Mid-latitude Summertime Mesoscale Convective Systems: a Simple Lagrangian Parcel Model Perspective*”.
- [39] **09/27/2023 Invited colloquium talk** in the Department of Meteorology and Atmospheric Science at the Pennsylvania State University. The title of my talk is “*Two theoretical frameworks on mid-latitude mesoscale convective systems over the Land*”.
- [38] **07/25/2023 Seminar talk** at the Atmospheric and Oceanic Sciences in the University of Wisconsin-Madison. The title of my talk is “*Impact of Global Warming on U.S. Summertime Mesoscale Convective Systems: A Simple Lagrangian Parcel Model Perspective*”.
- [37] **07/25/2023 Seminar talk** at the Department of Earth, Atmospheric and Planetary Sciences in the Purdue University. The title of my talk is “*Impact of Global Warming on U.S. Summertime Mesoscale Convective Systems: A Simple Lagrangian Parcel Model Perspective*”.
- [36] **07/17/2023 Oral talk** at the AMS 20th Conference on Mesoscale Processes in Madison, Wisconsin. The title of my talk is “*Impact of Global Warming on U.S. Summertime Mesoscale Convective Systems: A Simple Lagrangian Parcel Model Perspective*”.
- [35] **05/17/2023 Seminar talk** at the Center for Atmosphere Ocean Science (CAOS) Colloquium in New York University. The title of my talk is “*Impact of Global Warming on U.S. Summertime Mesoscale Convective Systems: A Simple Lagrangian Parcel Model Perspective*”.
- [34] **05/11/2023 Seminar talk** at the Atmospheric Dynamics Meetings at Princeton University and Geophysical Fluid Dynamics Laboratory (GFDL). The title of my talk is “*Impact of Global Warming on U.S. Summertime Mesoscale Convective Systems: A Simple Lagrangian Parcel Model Perspective*”.
- [33] **04/21/2023 Seminar talk** at the Atmospheric and Oceanic Sciences Colloquium (ATOC) in University of Colorado Boulder. The title of my talk is “*Impact of Global Warming on U.S. Summertime Mesoscale Convective Systems: A Simple Lagrangian Parcel Model Perspective*”.

- [32] **04/20/2023** Seminar talk at the Mesoscale & Microscale Meteorology (MMM) Laboratory of NCAR. The title of my talk is “*Impact of Global Warming on U.S. Summertime Mesoscale Convective Systems: A Simple Lagrangian Parcel Model Perspective*”.
- [31] **04/13/2023** Seminar virtual talk in the Meteorology seminar at the department of earth, ocean, & atmospheric science in Florida State University. The title of my talk is “*Impact of Global Warming on U.S. Summertime Mesoscale Convective Systems: A Simple Lagrangian Parcel Model Perspective*”.
- [30] **04/10/2023** Seminar talk in the Atmospheric & Chemistry Seminar at the department of atmospheric sciences, University of Washington. The title of my talk is “*Impact of Global Warming on U.S. Summertime Mesoscale Convective Systems: A Simple Lagrangian Parcel Model Perspective*”.
- [29] **12/15/2022** Oral talk in AGU fall meeting session “A45H: Understanding and Modeling of Mesoscale and Severe Local Convective Storm Processes III Oral”. The title of my talk is “*Impact of Global Warming on U.S. Summertime Mesoscale Convective Systems: A Simple Lagrangian Parcel Model Perspective*”.
- [28] **12/14/2022** Oral talk in AGU fall meeting session “A33E: High-Resolution Earth System Modeling on Large Supercomputers I Oral”. The title of my talk is “*Convective Momentum Transport and its Impact on the Madden-Julian Oscillation in E3SM-MMF*”.
- [27] **06/13/2022** **Invited virtual talk** in Canadian Applied And Industrial Mathematics Society (CAIMS) annual meeting, MS14 session: Models for the Atmosphere, Climate, and Ocean Dynamics. The title of my talk is “*A Simple Lagrangian Parcel Model for the Initiation of Summertime Mesoscale Convective Systems over the United States*”.
- [26] **01/25/2022** Remote poster presentation in the AMS’s 10th Symposium on the Madden-Julian Oscillation and Sub-Seasonal Monsoon Variability of the AMS 102nd Annual Meeting. The title of my talk is “*Upscale Impact of Mesoscale Convective Systems on the Madden-Julian Oscillation and Its Parameterization in a Coarse-Resolution GCM*”.
- [25] **01/24/2022** Remote presentation in the AMS’s 19th Conference on Mesoscale Process of the AMS 102nd Annual Meeting. The title of my talk is “*A Simple Lagrangian Parcel Model for the Initiation of Summer-time Mesoscale Convective Systems over the Central United States*”.
- [24] **12/14/2021** Virtual talk in the 2021 AGU fall meeting. The title of my talk is “*A Simple Lagrangian Parcel Model for the Initiation of Summer-time Mesoscale Convective Systems over the Central United States*”.
- [23] **12/13/2021** Virtual talk in the 2021 AGU fall meeting. The title of my talk is “*Upscale Impact of Mesoscale Convective Systems on the Madden-Julian Oscillation and Its Parameterization in a Coarse-Resolution GCM*”.
- [22] **09/02/2021** Virtual talk in the Climate Extremes monthly meeting. The title of my talk is “*A Simple Lagrangian Parcel Model for the Initiation of Summer-time MCSs over the Central US*”.
- [21] **05/11/2021** Virtual talk in the 34th Conference on Hurricanes and Tropical Meteorology in New Orleans, LA. The title of my talk is “*Upscale Impact of Mesoscale Convective Systems on the Madden-Julian Oscillation and Its Parameterization in a Coarse-Resolution GCM*”.
- [20] **01/08/2020** Seminar talk at the Institute of Natural Sciences, Shanghai Jiao Tong University. The title of my talk is “*Upscale Impact of Mesoscale Convective Systems on the Madden-Julian Oscillation and Its Parameterization in a Coarse-Resolution GCM*”.
- [19] **01/03/2020** Seminar talk at the Atmosphere-Ocean Department, Peking University. The title of my talk is “*Upscale Impact of Mesoscale Convective Systems on the Madden-Julian Oscillation and Its Parameterization in a Coarse-Resolution GCM*”.

- [18] **01/02/2020** Seminar talk at the Institute of Atmospheric Physics, Chinese Academy of Sciences. The title of my talk is “*Upscale Impact of Mesoscale Convective Systems on the Madden-Julian Oscillation and Its Parameterization in a Coarse-Resolution GCM*”.
- [17] **12/27/2019** Seminar talk at the Atmosphere-Ocean Department, Fudan University. The title of my talk is “*Upscale Impact of Mesoscale Convective Systems on the Madden-Julian Oscillation and Its Parameterization in a Coarse-Resolution GCM*”.
- [16] **12/11/2019** AGU fall meeting at San Francisco, poster presentation entitled “*Upscale Impact of Mesoscale Convective Systems on the Madden-Julian Oscillation and Its Parameterization in a Coarse-Resolution GCM*”..
- [15] **07/25/2019** **Invited talk** in the celebration conference in honor of Professor Andrew J. Majda’s 70th birthday, “Scientific Grand Challenges and New Perspectives in Applied Mathematics: Ocean, Atmosphere and Climate Sciences” at University of Victoria. Oral talk entitled “*Upscale Impact of Mesoscale Convective Systems on the CCEWs and MJO and Its Parameterization in an Idealized GCM*”.
- [14] **12/21/2018** 2018 Young Mathematician Forum at Department of Mathematical Sciences, Shanghai Jiao Tong University, China, Oral talk entitled “*Multi-scale interactions of organized tropical convection: from multi-scale asymptotic models to comprehensive numerical simulations*”.
- [13] **12/19/2018** Academic Colloquium at Department of Atmospheric and Oceanic Sciences, Fudan University, China, Oral talk entitled “*Upscale Impact of Mesoscale Convective Systems and Its Parameterization in an Idealized GCM for a MJO Analog above the Equator*”.
- [12] **12/10/2018** AGU fall meeting at Washington, DC, poster presentation entitled “*Upscale Impact of Mesoscale Convective Systems and Its Parameterization in an Idealized GCM for a MJO Analog above the Equator*”.
- [11] **05/03/2018** **Invited talk** in Climate & Global Dynamics (CGD) research report at NCAR, Oral Talk entitled “*Upscale Impact of Mesoscale Convective Systems and Its Parameterization in GCMs: an idealized testbed*”.
- [10] **04/16/2018** American Meteorological Society’s 33rd Conference on Hurricanes and Tropical Meteorology, Oral Talk entitled “*Upscale Impact of Mesoscale Disturbances of Tropical Convection on Convectively Coupled Kelvin Waves*”.
- [9] **12/11/2017** AGU fall meeting at New Orleans, Poster Presentation.
- [8] **11/29/2017** Applied math seminar in Department of Mathematics and Statistics at UVic, Victoria, Canada, Oral Talk.
- [7] **05/09/2017** Seminar talk at Department of Atmosphere Science, Princeton University, New Jersey, Oral Talk.
- [6] **12/12/2016** AGU fall meeting at San Francisco, Poster Presentation.
- [5] **09/15/2016** AOCD seminar series at Department of Geology & Geophysics, Yale University, New Haven, Oral Talk.
- [4] **01/28/2016** Multidisciplinary University Research Initiative (MURI) Workshop, New York, Oral Talk.
- [3] **09/01/2015** Columbia Workshop “Monsoons & ITCZ: the annual cycle in the Holocene and the future”, New York, USA, Poster Presentation.
- [2] **04/01/2015** BIRS Workshop “Stochasticity and Organization of Tropical Convection”, Banff, Canada, Oral Talk.

- [1] **08/16/2014** The World Weather Open Science Conference (WWOSC), Montreal, Canada, Poster Presentation.

HONORS

Outstanding Faculty Mentor Award for Undergraduate Research Training, Peking University	2025
Second Prize in the Science and Engineering Category and the Outstanding Teaching Plan Award at the 24th Young Teachers' Teaching Skills Competition of Peking University	2025
Peking University WeiMing Young Scholar	2024
XieYiBing Award for Young Scholars in Meteorology and Technology	2024
NYU GSAS Dean's Student Travel Grant Program	2016
McCracken scholarship of New York University	2012-2017
Meritorious Award in the Mathematical Contest in Modeling	2010

TEACHING

Instructor for International Earth Science Olympiad 2025 National Team Training	July 2025
Undergraduate Course:	
Atmospheric Dynamics	Spring 2024
Numerical Simulation and AI Forecast for Geophysics Fluids (I set up this course)	Fall 2024
Graduate Course:	
Mesoscale Dynamics	Fall 2024
Advances in Atmospheric Sciences	Fall 2024
Recitation Leader for Algebra and Calculus, MATH-UA.9.002-003	Spring 2015
(Overall teaching score based on students' evaluation = 4.0/5.0)	

MENTORING

PhD student:
 Huaiping Wang (2024 Fall), Haobo Yang (2025 Fall)

Undergraduate student:
 2022 Class: Yuhui Li, Wencan Zhu, Zeyu Tang
 2023 Class: Dazhuang Cao, Yifan Zhao

COMMUNITY SERVICE

Junior Editorial Board Member for Meteorological Monthly	since 07/08/2026
Associate Editor for Quarterly Journal of the Royal Meteorological Society	Since 10/01/2025
Associate Editor for Journal of the Atmospheric Sciences	Since 09/01/2022
Editorial Board Member for Journal of Meteorological Research	since 04/09/2025
Member of climate and environment mathematical committee in China Society for Industrial and Applied Mathematics	Since 10/01/2025
Member of atmospheric science committee in Chinese Meteorology Society	Since 10/01/2024
Member of urban meteorology committee in Chinese Meteorology Society	Since 10/01/2024
Member of dynamic meteorology committee in Chinese Meteorology Society	Since 10/01/2024
Member of atmospheric science terminology committee in Chinese Meteorology Society	Since 10/01/2024
Reviewer for Banff international research station workshop proposal	Sep 2022
Reviewer for U.S. NSF proposals	Jun 2023
Paper reviewer for	since 2017
<i>Journal of the Atmospheric Sciences</i>	
<i>Journal of Climate</i>	
<i>Climate Dynamics</i>	

Dynamics of Atmospheres and Oceans
Journal of Geophysical Research: Atmospheres
Quarterly Journal of the Royal Meteorological Society
International Journal of Climatology
npj climate and atmospheric science
Journal of Advances in Modeling Earth Systems
Theoretical and Applied Climatology
Advances in Atmospheric Sciences (AAS)
Atmosphere
 Chair of 2022 AGU session “A41E: Understanding and Modeling of Mesoscale and Severe Local
 Convective Storm Processes I Online Poster Discussion” Dec 2022
 Chair of 2018 AGU session “A13K: Panel discussions: Clouds in Tropics: Stochasticity and
 Organization, Observation, and Simulation” Dec 2018
 Organizing Chair of Young Scientist Professional Development (YSPD) group at PNNL 2022
 Outstanding Student Presentation Award (OSPA) Judge for AGU fall meeting Since 2019
 Member of the reviewer board & topic board for the MDPI journal Atmosphere since Nov 2019
 Member of American Meteorological Society and American Geophysical Union since 2016
 Research proposal reviewer for National Science Center, Poland June 2017

COMPUTER SKILLS

Programming languages:

Fortran, C++, MATLAB, Python, Linux, High-Performance Computing clusters, Latex, Lyx

Running and revising codes for models:

System for Atmospheric Modeling (SAM), Weather Research and Forecasting Model (WRF),
 HOMME dynamic core for CAM, both deterministic and stochastic multcloud models

LANGUAGE

English (fluent), Mandarin (native)

REFERENCES

[1] **Andrew J. Majda** (my Ph.D. advisor)

Title: Member of US National Academy of Sciences, Morse Professor of Arts and Science at the Courant
 Institute of New York University

Affiliation: Center for Atmosphere and Ocean Science, Courant Institute of Mathematical Sciences,
 New York University, USA

Email: jdm11@cims.nyu.edu (This email belongs to his secretary Jacquelyn Mileski)

[2] **Lai-yung Ruby Leung** (principal investigator of my current projects)

Title: Member of US National Academy of Engineering, Battelle Fellow at Pacific Northwest National
 Laboratory, Fellow of American Meteorological Society, Fellow of American Geophysical Union

Affiliation: Atmospheric Science and Global Change Division at Pacific Northwest National Laboratory,
 USA

Email: ruby.leung@pnnl.gov

[3] **Mitchell W. Moncrieff** (one of my Postdoc mentors)

Title: NCAR Senior Scientist, Fellow of American Meteorological Society, Fellow of Royal Meteorological
 Society

Affiliation: Climate and Global Dynamics Laboratory, National Center for Atmospheric Research, USA

Email: moncrief@ucar.edu

[4] **Boualem Khouider** (one of my Postdoc mentors)

Title: Professor, Editor of the Journal “Mathematics of Climate and Weather Forecasting”

Affiliation: Department of Mathematics and Statistics, University of Victoria, Canada

Email: khouider@uvic.ca